Case Study 4 - NATA Supermarkets: Customer Analytics

Mark-6580-400, Prof. John Gale

Saxa Group 3 Students:

- Dylan Lowndes
- Emma Cranmer
- Israel Mendoza
- Mike Johnson
- Lola Oshodi

April 17, 2025

1. *What steps should be taken to clean the data?*

We approached data cleaning as a crucial step to ensure the reliability and accuracy of the insights we derived. Some general steps that NATA Supermarkets should take include:

- **Identifying and handling missing values**: NATA's data set included 24 null values in 'Income'. To handle these, NATA could either impute the mean income or remove these records from the analysis. For this analysis, these nulls were removed.
- **Detecting and resolving inconsistencies**: This involves standardizing data formats and ensuring consistent categorical data types. NATA's dataset had inconsistent date formats in 'DT_Customer' that were transformed to a consistent date-time format. In addition, 'Marital_Status' had several inconsistent categories (Figure 4). These values were updated to "Single" for analysis.
- **Identifying and handling outliers**: Outliers can skew analysis and need to be investigated and potentially removed or transformed. NATA's dataset had outliers detected in 'Year_Birth' (Figure 5) and 'Income' (Figure 6). These outliers were removed.

2. Perform a descriptive analysis of the data and visualizations relevant to the problem.

NATA's dataset captured the shopping histories of 2,240 customers from 2012 to 2014. After removing missing values and outliers, 2,205 customers remained for analysis. Most of these customer households had annual incomes between $35,000 and $68,000, with a median income of $51,287. Customer ages primarily clustered in the mid-50s to mid-60s, averaging around 56 years old, indicating that NATA's core shoppers were nearing or already in retirement. On average, customers returned to shop every 49 days, although actual intervals ranged widely from same-day visits to gaps of nearly three months.

- **Customer Income:** Customer income was identified as a key variable, given its direct impact on purchasing power. Income across the dataset ranged from $1,730 to $113,734, with an average of $52,622. The primary target for predictive modeling was customers' total spending, calculated as the sum of all product purchases over two years, which ranged from $5 to $2,525, with an average spend of $606. Most customer spending, however, fell in the lower range, between $0 and $100.
- **Spending Patterns:** Key findings highlighted spending patterns, particularly the dominance of premium product categories like wine and meat, which accounted for approximately 75% of total consumer spending over the two years (Figure 3). Even small improvements in these categories could significantly increase overall revenue.
- **Household Composition:** Additionally, customers shopping as couples ("Married" or "Together") spent roughly 15–20% more than single customers, suggesting potential benefits in promoting "date-night" or family-focused product bundles.

3. What models are appropriate to predict customer spending?

While a variety of models can be used to predict customer spending based on the attributes in the dataset, including linear regression, decision trees, neural networks, and random forests, we selected linear regression as the most appropriate for this analysis (see code below) because of the following reasons:

Linear regression is particularly valuable due to its interpretability, which is essential for this project. Since the goal is to generate actionable insights and recommendations for the NATA Supermarket leadership team, it is crucial to clearly understand the impact of each attribute on customer spending. Unlike neural networks and random forests, which often function as "black box" models with limited transparency, linear regression provides clear and interpretable relationships between input features and the predicted outcome of a customer's spend, such as total spend by customer. These insights can be used to generate specific marketing tactics, where the marginal effect can also be estimated with the model.

Additionally, linear regression is well-suited for the size of our dataset, which consists of just over 2,200 observations. While neural networks can be highly effective with large datasets, they tend to underperform with smaller data samples. In contrast, linear regression delivers reliable performance and meaningful insights, making it the most practical and informative choice for this analysis. This will also increase the speed at which teams at NATA can refresh and iterate on the models to provide timely insights to business leadership to make key decisions with more agility.

Model Results and Interpretation

The model on test data had an R-squared value of 0.76 (Figure 7), meaning that the regression model explains 76% of the variability in the total purchases. Interpretation of the model's coefficients:

- *Income (+0.01)* - Total purchases increase by $0.01 for every $1 increase in Income.
- *High Income (664)* - High-income customers, on average, spend $664 more than the low-income segment.

The full regression code used to train and evaluate this model can be found in **Figure 9**.

4. What models are appropriate to segment customers?

When selecting a model for customer segmentation, several approaches can be used, such as K-means or hierarchical clustering. Among these, K-means clustering is the most widely used and recommended model for this analysis (see code below). Much like the advantages of linear regression, K-means provides a high level of flexibility, as it allows us to define the number of customer segments based on the specific business needs.

One of the key benefits of K-means is its interpretability. Being able to clearly explain how the segmentation model works and which attributes drive the clustering decisions will be crucial in making actionable recommendations to NATA's leadership team. With a range of customer data, such as demographic information (e.g., education, age, marital status) and behavioral characteristics (e.g., spending patterns), K-means allows us to easily highlight the key factors that contribute to different customer groups. This is vital for tailoring marketing strategies or other business decisions.

Another major advantage of K-means is its speed and scalability. It is an agile solution that allows for rapid iteration and fine-tuning as new data becomes available. This is particularly useful when trying to derive actionable insights quickly. In contrast, models like hierarchical clustering are often slower and more rigid, requiring more computing power, especially with larger datasets, and making them less flexible for frequent adjustments.

K-Means Results

Cluster analysis identified two distinct groups of customers:

- **Low Income (Cluster 0)** - This segment has a mean income of $38K**.** The lower income translates to a lower amount of purchases across all categories.
- **High Income (Cluster 1)** - Average income in this group is $71K and sees a higher amount of purchases overall.

K-Means Code

```
optimal_k = 2 # Adjust this number based on the elbow plot result

kmeans = KMeans(n_clusters=optimal_k, random_state=206)

df_ml['cluster'] = kmeans.fit_predict(df_ml_scaled)
```

5. What actionable insights can be provided to the retailer?

The analysis reveals a clear link between income and customer spending, with high-income customers spending significantly more, on average $664 above their lower-income counterparts. This insight, combined with the identified customer segments, presents a strong case for Nata to move away from broad, uniform marketing strategies and toward more targeted, data-driven initiatives. However, income alone may not capture the full complexity of customer behavior—factors like family size, lifestyle preferences, or even product availability could influence purchasing patterns in ways that aren't fully reflected in the model. Still, the strength of the correlation between income and spending, alongside clear segment-level trends, provides a solid foundation for designing differentiated strategies that align more closely with customer needs and potential value.

For the high-income segment, NATA should elevate its value proposition by promoting premium products and creating exclusive shopping experiences. Bundled offerings—such as curated wine and meat packages—can increase basket size while reinforcing NATA's brand as a quality-focused retailer. Personalized loyalty programs and digital outreach can further enhance engagement, offering incentives aligned with their preferences and lifestyle.

In contrast, the low-income segment responds more to value and savings. Here, NATA can focus on essential product discounts, clear in-store messaging, and simple promotions tied to household needs, like "meals under $20." A gamified loyalty program or rewards tied to spending thresholds can encourage incremental purchases while fostering a sense of inclusion and reward.

Across segments, these strategies should be reinforced by improved inventory alignment and real-time campaign testing. By tailoring promotions, pricing, and product mix to the distinct behaviors of each customer segment we identified, NATA can drive both customer satisfaction and business performance.

Appendix

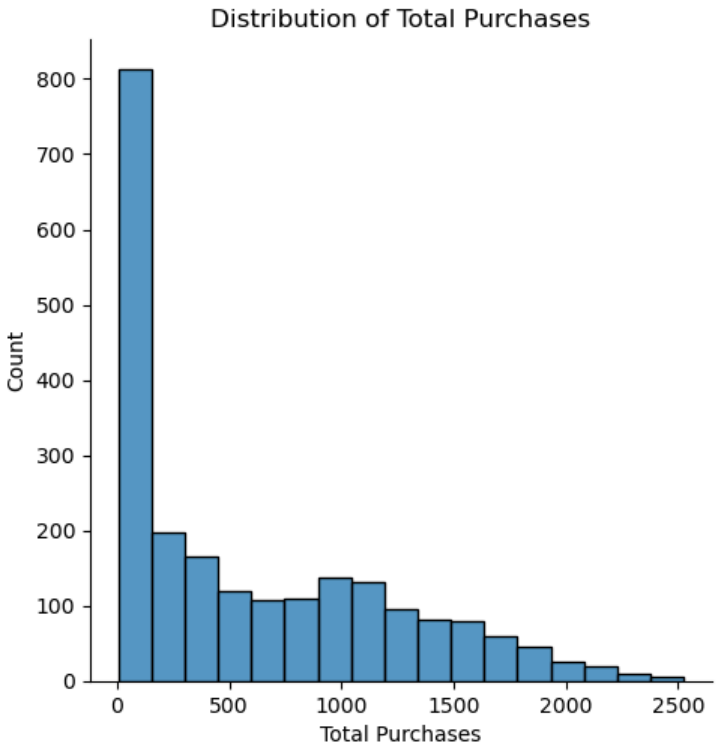

Figure 1: Distribution of Total Purchases

This histogram shows the distribution of total customer purchases. Most customers spent under $500 over two years, with a significant concentration around $0–$100. Only a small group of customers spent above $1,000, highlighting opportunities for increasing average customer spend.

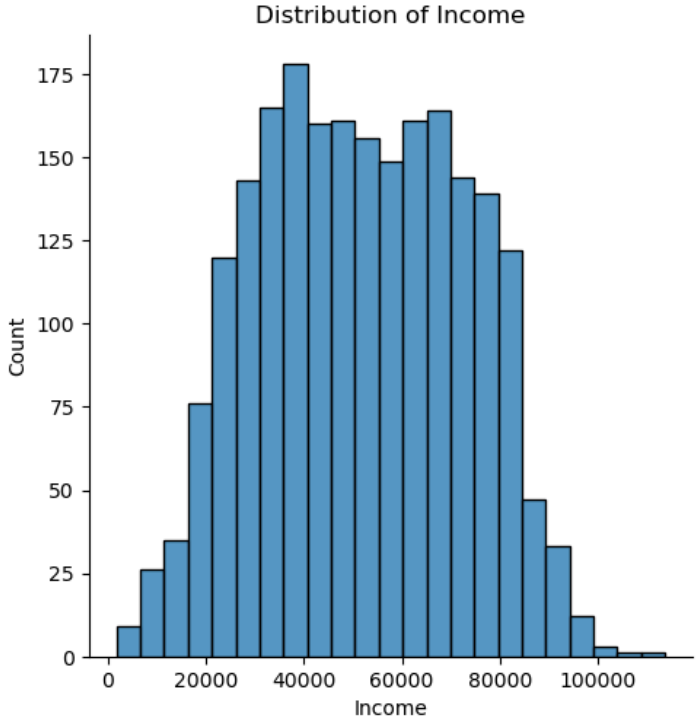

Figure 2: Distribution of Income

The histogram illustrates the income distribution of NATA's customers. Incomes primarily range between $35,000 and $68,000, with fewer customers at very low or very high income levels. This income pattern is crucial for targeted marketing and pricing strategies.

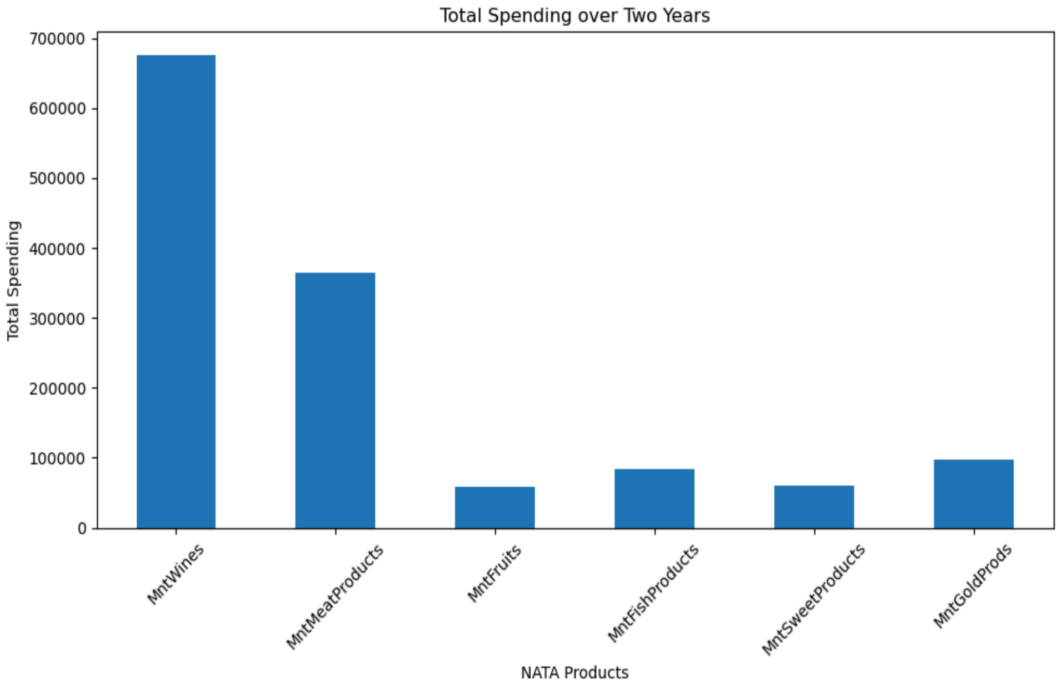

Figure 3: Total Spending by Product

The chart shows where customers spent the most over a two year period. Wine and meat products make up most of the total spending. This highlights the premium categories and marketing opportunities.

Marital_Status	**Count**
Absurd	2
Alone	3
Divorced	230
Married	854
Single	470
Together	568
Widow	76
YOLO	2

Figure 4: **Data Inconsistencies in 'Marital_Status'**

The table highlights inconsistent categories found in the original dataset's "Marital_Status" field, i.e "Absurd," "Alone," and "YOLO." These irregular categories were consolidated into the "Single" category to better streamline the analysis.

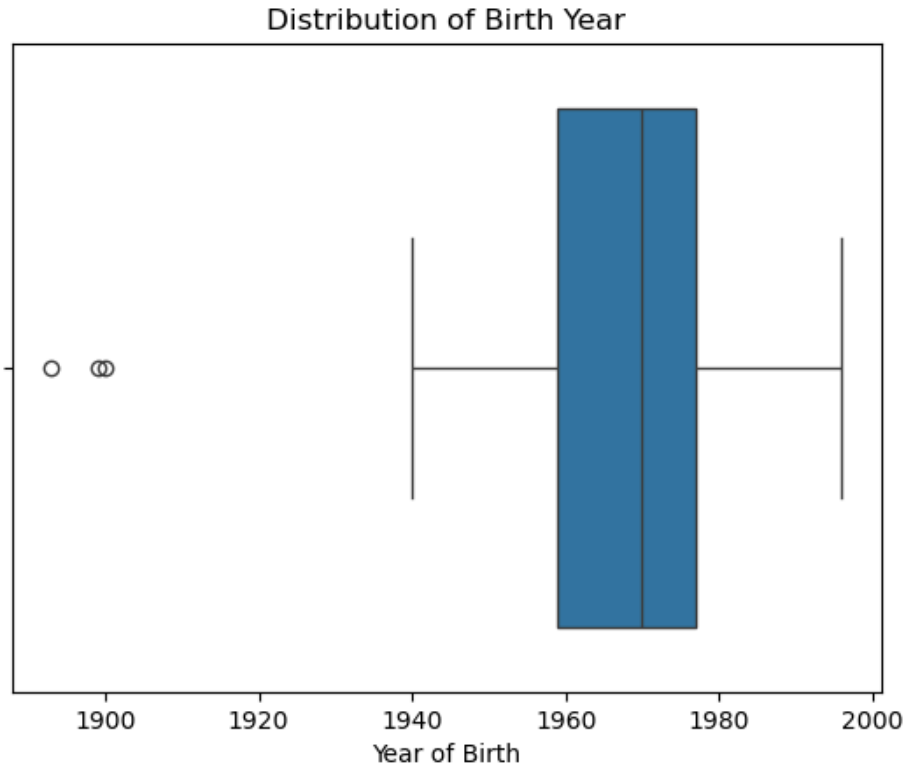

Figure 5: Outlier Detection - Birth Year

The boxplot helped to identify a number of birth years outliers that were significantly higher than the typical customer. This suggests entry errors. These outliers were removed to give a better customer age profile.

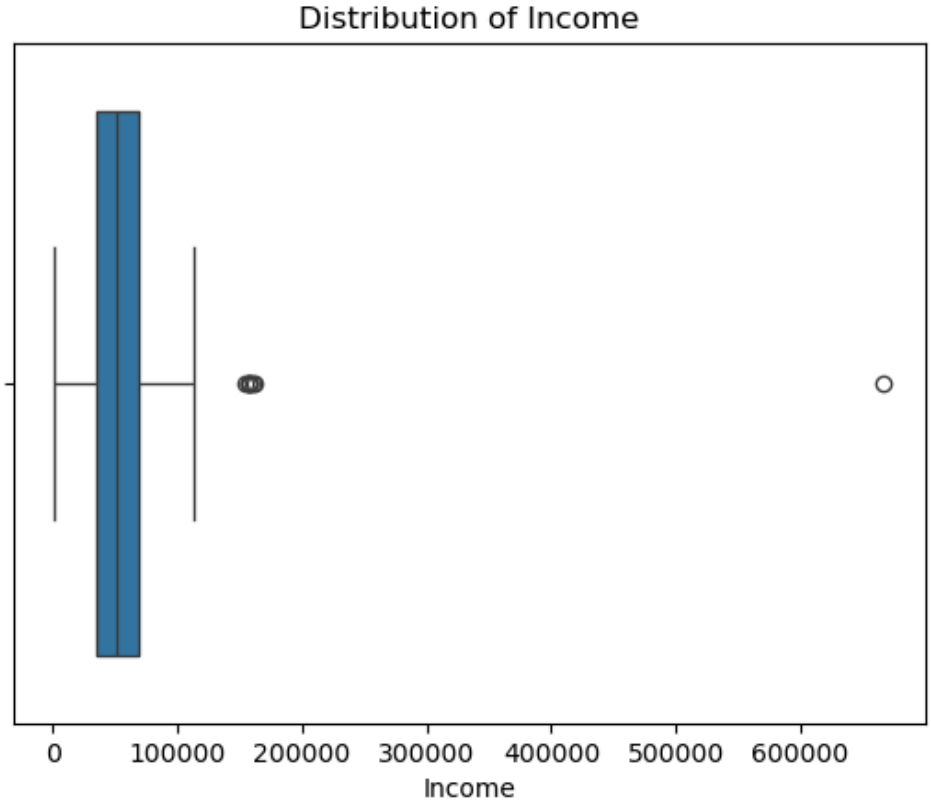

Figure 6: Outlier Detection - Income

The box plot helped show extremely high incomes. Any significant outliers (incomes) were removed so the analysis could reflect the spending patterns of the general customer base.

Figure 7: Linear Regression Model Performance on Test Data

Mean Squared Error (MSE): 78644.19

Root Mean Squared Error (RMSE): 280.44

Mean Absolute Error (MAE): 207.61

R-squared (R2): 0.76

The linear regression model did a good job predicting total purchases. An R-squared value of 0.76, means the model accounts for about 76% of the differences in how much customers spend. The model's RMSE of 280.44 and MAE of 207.61 indicate reasonable prediction error, making it a solid choice for interpreting how key features like income, and age and how they relate to consumer spending.

Figure 8: Segmentation by Income and Total Purchases

This scatterplot shows two clear customer segments: higher-income shoppers (orange), earning mostly over $60,000 and spending significantly more, and lower-income shoppers (blue), generally earning below $60,000 with lower total spending. This highlights how income strongly influences spending habits, guiding targeted marketing efforts.

Figure 9: Linear Regression Code

```
# Create total_purchases

df_ml['total_purchases'] = df['MntWines'] + df['MntFruits'] + df['MntMeatProducts'] +
df['MntFishProducts'] + df['MntSweetProducts'] + df['MntGoldProds']
```

```
# Drop purchase columns

df_ml = df_ml.drop(['MntWines', 'MntFruits', 'MntMeatProducts', 'MntFishProducts',
'MntSweetProducts', 'MntGoldProds'], axis=1)

        # Define y variable

        y = df_ml['total_purchases']

        # Define x variables

        X = df_ml.drop('total_purchases', axis = 1)

        # Apply train test split

        X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state =
        206)

        # Update X_train with just cluster and income

        X_train = X_train[['Income', 'cluster_1']]

        X_test = X_test[['Income', 'cluster_1']]

        # Prelim model

        lm_1 = LinearRegression()

        lm_1.fit(X_train, y_train)

        # Define cross-validation strategy

        cv = KFold(n_splits = 10, shuffle = True, random_state = 206)

        # Perform cross-validation and get the scores (R-squared by default)

        scores = cross_val_score(lm_1, X_train, y_train, cv=cv, scoring='r2')

        # Model Results

        y_pred = lm_1.predict(X_test)

        mse = mean_squared_error(y_test, y_pred)

        rmse = np.sqrt(mse)

        mae = mean_absolute_error(y_test, y_pred)

        r2 = r2_score(y_test, y_pred)
```